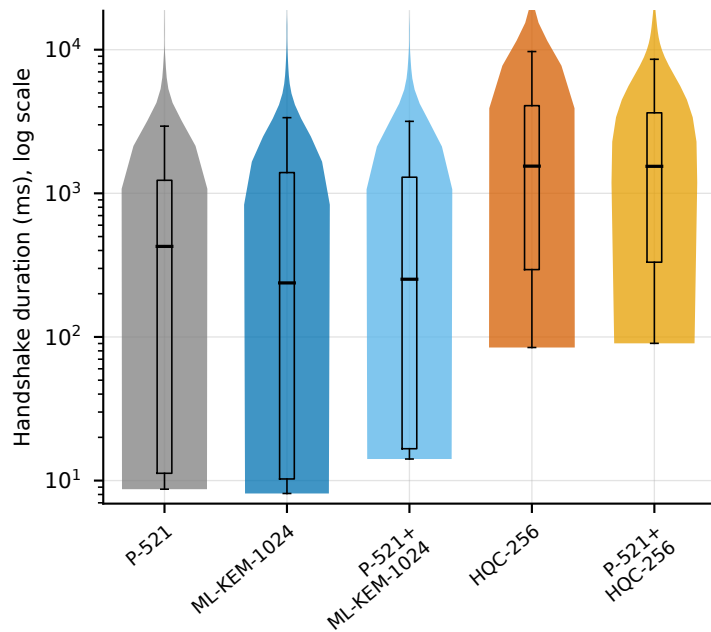
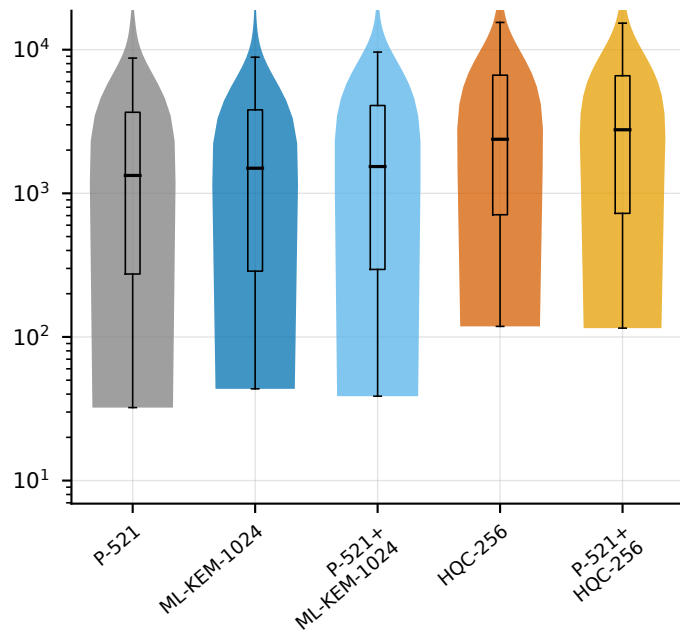


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)	mldsa87 Fail
P-521	1.40k	403.6	105.44k	1/500	4.57k	269.8	113.24k	1/500
ML-KEM-1024	1.53k	346.9	81.72k	2/500	4.48k	239.5	108.43k	1/500
P-521+ML-KEM-1024	1.57k	368.7	104.07k	1/500	5.59k	271.3	112.80k	5/500
HQC-256	5.00k	392.8	378.93k	5/500	8.11k	212.4	133.79k	7/500
P-521+HQC-256	4.56k	262.2	108.79k	3/500	7.67k	213.1	114.98k	5/500